Extracurriculars & Leadership

PATH by Rohin | RAG Knowledge Source: Depth, Leadership, Initiatives, Competitions, Research & Internships

Purpose and scope

This document provides general guidance for middle school and high school students planning extracurricular activities, leadership experiences, independent initiatives, academic competitions, research, and internships. It explains practical decisions and common misconceptions. It is not a school policy, an admissions guarantee, or a substitute for requirements published by a particular institution. Opportunities and eligibility rules vary by school, program, age, location, and application year. Students should verify requirements directly with the sponsoring organization and discuss substantial commitments with a parent, guardian, teacher, or counselor when appropriate.

What counts as an extracurricular activity?

An extracurricular activity is a meaningful commitment outside a student's required academic classes. Examples include school clubs, paid work, family responsibilities, community volunteering, arts, athletics, religious or cultural organizations, independent programming, writing, robotics, tutoring, advocacy, entrepreneurship, research, and caring for relatives. An activity does not have to be affiliated with a school or carry a prestigious title to count. A student who consistently helps run a family business or supervises siblings may be demonstrating significant responsibility even without a club officer position. The important questions are what the student actually did, why it mattered, what they learned, and how their responsibility changed over time.

Depth versus breadth

Breadth means exploring different interests; depth means developing sustained involvement, skills, responsibility, or impact within an interest. Early in high school, trying several activities can help students discover what they enjoy. As interests become clearer, continuing with a smaller set of meaningful commitments often produces richer learning than joining many clubs without participating. There is no universally correct number of activities and no formula that guarantees college admission. A student can show depth by practicing consistently, mentoring newcomers, completing challenging projects, organizing events, or improving an existing process. Depth does not require staying in an activity that is harmful, inaccessible, or no longer appropriate. Students may change direction as they learn more about themselves.

Choosing activities intentionally

Start with curiosity and practical constraints. List subjects and causes that interest you, available opportunities, transportation needs, family commitments, costs, and time for schoolwork and rest. Select a manageable combination of activities that lets you explore, contribute, and develop skills. Ask what you could learn in three months, what meaningful contribution you could make in a year, and whether you enjoy the work even without an award. An academic interest might lead to a coding club, science fair, independent research, or a library workshop. A service interest might lead to tutoring, food-bank volunteering, or organizing accessible educational resources. Interests can overlap, but a student's activities do not all need to match the intended college major.

How leadership is demonstrated

Leadership is taking responsibility for helping people or a project move toward a useful goal. Formal positions such as president, vice president, secretary, or team captain are one form of leadership, but students can also lead without a title. Examples include training new members, creating documentation, resolving a recurring problem, coordinating volunteers, organizing a workshop, and following through on commitments. Leadership should not be confused with claiming sole credit for group work. Students should describe their actual role and recognize the contributions of teammates, faculty advisers, and mentors. A title is most convincing when paired with specific actions and outcomes.

Earning leadership roles

Students interested in officer positions can begin by attending consistently, volunteering for tasks, completing work reliably, and listening to existing members. Learn the organization's mission and operating rules before proposing changes. Offer to maintain a website, coordinate communications, plan a meeting, mentor a new participant, or manage a small event. Keep track of results and ask for constructive feedback. When an election or application becomes available, explain what you have contributed and what you plan to improve. If not selected, continue contributing and seek responsibility in other ways. Students should not treat leadership elections as the only measure of growth.

Starting a club or independent initiative

Before starting a club, determine whether an existing organization already serves the same need. A new initiative is valuable when it addresses a real gap, creates a better way to serve people, or enables a distinctive project. Define a specific audience, a concrete problem, and a feasible first offering. For example, a student interested in programming might host a small beginner workshop before attempting to build a large educational nonprofit. At a school, obtain any required faculty sponsor, approval, meeting space, safety permissions, and accessibility accommodations. For a public website, consider content accuracy, privacy, moderation, and maintenance. Start with a pilot, collect feedback, revise, and document what actually happened. Founding an initiative is not the same as demonstrating impact; meaningful outcomes depend on execution.

Measuring impact without exaggeration

Impact may be quantitative or qualitative. Useful quantitative measures include participants taught, events completed, volunteer hours, devices distributed, resources published, project users, funds raised, or improvements in turnaround time. Useful qualitative evidence includes feedback from participants, a teacher's observations, examples of increased accessibility, or a problem solved for the community. Count only verifiable outcomes. Distinguish planned goals from completed work: 'developing a tutoring platform' is not the same as 'tutored 200 students.' If a project has a small audience, describe the actual work rather than inflating its scale. Protect privacy when collecting feedback or usage metrics, particularly when working with minors.

Competitions and academic challenges

Competitions can provide deadlines, collaboration, technical practice, and feedback. Options vary widely: robotics, coding, mathematics, debate, business, science fairs, writing, arts, and community innovation challenges. Choose competitions aligned with genuine interests and available resources, not only the perceived prestige of an award. Read the official eligibility rules, submission format, originality requirements, deadlines, and restrictions on outside assistance or AI tools. In a team competition, define roles clearly and document individual contributions. Not winning does not make the experience worthless: the design process, prototype, debugging, presentation, and reflection can all demonstrate growth. Report awards accurately, including the level of competition and whether a result was a nomination, placement, finalist designation, or win.

Research for high school students

Research begins with a question that can be investigated using evidence and a suitable method. A project may involve literature review, data collection, experiments, model development, analysis, and interpretation. Students should start with a narrow, answerable question rather than a vague goal such as 'research AI.' A research mentor may suggest ideas, review methodology, critique drafts, and teach ethical practices while the student conducts substantial independent work. 'Mentored independent research' is a reasonable description when the student drives the execution and receives guidance; 'collaborative research' may be more accurate when others substantially co-design or co-produce the work. Be transparent about contributions and authorship. Publishing is not required for research to be meaningful, and a planned submission must never be described as an accepted publication.

Research integrity and responsible methods

Keep reproducible records of data sources, permissions, experimental procedures, code versions, and changes in methodology. Understand the difference between training data and test data, and avoid evaluating a model only on data it has already seen. Describe uncertainty, limitations, and results that do not support the initial hypothesis. Never fabricate findings, citations, or outcomes. Projects involving human participants, sensitive personal information, biological materials, or potentially hazardous experiments may require special supervision, ethics review, or school approval before work begins. Mentors and school staff can help determine applicable rules. Public datasets and code repositories also have licensing and privacy requirements.

Finding internships and professional experiences

An internship is a structured opportunity to learn and contribute in a professional setting; paid work, job shadowing, volunteer projects, and apprenticeships may offer similar learning. Students can explore school career programs, local businesses, community organizations, university outreach, and carefully vetted remote opportunities. Prepare a concise portfolio of actual work and a short message describing relevant skills, interests, availability, and the kind of contribution you hope to make. A strong internship is not defined solely by a company name. Consider supervision quality, safety, accessibility, meaningful responsibilities, and whether the experience teaches useful professional habits. Do not pay suspicious placement fees or share sensitive information with unverified recruiters. Minors should review employment rules and arrangements with a parent or guardian.

Making the most of an internship

Set realistic learning goals, clarify responsibilities, communicate progress, and ask for feedback. Learn how teammates document requirements, review code, test systems, and handle mistakes. Respect confidentiality and intellectual-property agreements; a student should not publish employer code or internal materials without permission. When describing internship work publicly, distinguish 'built independently,' 'contributed to,' and 'worked with a team on.' Explain the problem, your contribution, the tools used, and what you learned. An invitation to return can be mentioned if it actually occurred, but should not substitute for explaining the substance of the work.

Balancing academics, activities, and well-being

Extracurricular involvement should be sustainable. Estimate weekly hours honestly, including commuting, meetings, preparation, and recovery time. Keep room for sleep, coursework, friendships, and unstructured time. During exam periods, reduce optional commitments when possible and communicate early with teammates. If several activities overlap, consider which ones provide the most meaning and where your responsibilities are greatest. Students should not pursue exhaustion as evidence of ambition. A smaller number of well-managed commitments can be more valuable personally than a crowded schedule that undermines learning or health.

Planning by grade level

Middle school: explore interests, build foundational skills, and try accessible clubs or service opportunities without treating every choice as a college application strategy. Ninth grade: join activities, learn school expectations, and notice which interests motivate continued effort. Tenth grade: deepen participation, complete tangible projects, and begin taking on responsibilities. Eleventh grade: pursue more advanced work, mentorship, leadership, or research where appropriate while protecting academic performance. Twelfth grade: follow through on existing commitments, mentor younger students, and report completed achievements accurately. These are flexible suggestions, not required milestones. A student can begin a meaningful activity at any age or grade.

Documenting activities for applications and recommendations

Maintain a simple activity record with the organization's full name, position, dates, approximate hours per week, weeks per year, responsibilities, outcomes, and an adult contact who can verify participation if needed. Record awards separately with their exact titles and levels. For college applications, use the allotted description space to show actions and contributions rather than repeating the organization name. Quantify results when verifiable, and distinguish ongoing work from future plans. Recommendation writers benefit from specific examples they personally observed, such as initiative, persistence, teamwork, mentoring, or a project that improved over time. Do not fabricate numbers to fill a character limit.

Common questions and answers

Question: Is joining ten clubs better than joining three? Answer: Not inherently. Participation quality, genuine interests, sustained work, and contributions matter more than the number alone. Question: Do I need to be club president? Answer: No. Leadership can be demonstrated through responsibility and initiative without a formal title. Question: Can personal coding projects count? Answer: Yes, independent projects can be extracurricular activities when they involve meaningful work outside class. Question: Does research need to be published? Answer: No; rigorous methodology, honest analysis, and learning matter even without publication. Question: What if my school offers few clubs? Answer: Colleges can consider opportunity context; students can explore work, family responsibilities, community activities, or feasible independent projects. Question: Can I describe a project that is still being built? Answer: Yes, if its current status is stated honestly and plans are not represented as completed impact.

Core guidance for students

Explore broadly enough to discover genuine interests, then invest deeply where your effort can grow into skill, responsibility, and contribution. Seek leadership through service rather than titles alone. Build or start something when you see a real need, test it with a manageable pilot, and improve it based on evidence. Treat competitions, research, and internships as learning experiences, not guaranteed admissions advantages. Keep accurate records, credit collaborators, respect ethical boundaries, and balance activities with academics and well-being. Every student's circumstances differ, so an effective pathway is one that is authentic, sustainable, and suited to the opportunities available.

Extracurricular Examples and Student Pathways

Supplement to PATH by Rohin - Extracurriculars & Leadership

Examples of Extracurricular Activities Across Interests

Extracurricular activities are sustained pursuits outside required coursework. Examples include school clubs, independent projects, caregiving, part-time jobs, athletics, arts, community organizing, research, and volunteering. An activity does not need a prestigious title, competition award, or national affiliation to matter. Strong examples often show genuine interest, increasing responsibility, specific contributions, reflection, and continuity over time. Students admitted to selective colleges have pursued many different activities; there is no universally successful activity checklist, and admission cannot be guaranteed by copying another student.

The following examples are illustrative pathways, not a ranking of activities or a claim that every example has led to a particular admission result. Students should choose activities they can access and care about, rather than attempting to imitate an impressive-sounding resume. The examples from Rohin are personal portfolio examples, not requirements or a formula for admission.

Computer Science and Artificial Intelligence: Project and Builder Examples

A student interested in computer science might build a homework planner, accessible school club website, neighborhood volunteer scheduling tool, language-learning application, educational game, or data visualization for a local nonprofit. A beginner can start with one working feature, document the problem, ask intended users for feedback, improve reliability, and publish code with a clear README. Useful evidence includes working demonstrations, design decisions, version history, user feedback, and accessibility improvements.

A student interested in machine learning might classify text sentiment, compare forecasting models, build an image classifier, or create a retrieval-augmented generation chatbot that answers questions using a curated document library. For a trustworthy AI project, the student should describe data provenance, model limitations, evaluation methods, privacy precautions, and the difference between a prototype and a deployed tool. An AI model does not automatically create community impact; that requires evidence that people used or benefited from the tool.

Rohin portfolio example - AI internship: Rohin worked with professional AI and machine learning engineers at Ejyle Technologies on a retrieval-augmented generation system. His reported stack included Ollama, Qdrant for dense and sparse retrieval, BGE-Large embeddings, and OpenTelemetry/OpenInference with Arize Phoenix for observability. The educational lesson is that technical work can progress from classroom coding to collaborative engineering, testing, and evaluation. This example does not imply that every high school student needs an internship.

Rohin portfolio example - PATH by Rohin: Rohin is developing a React and TypeScript student-guidance website and an Ask PATH chatbot informed by retrieval-augmented generation concepts from his internship. Its intended purpose is to help younger students navigate high-school information while separating verified school requirements from personal advice. As a developing project, its reach should

not be presented as established until measured. The transferable lesson is to reuse technical skills to solve a problem one understands personally.

Rohin portfolio example - stock sentiment: During InspiritAI, Rohin collaborated on a BERT-based natural-language processing project that classifies stock-related text sentiment as positive, negative, or neutral. He later independently developed a React/TypeScript interface and integrated hosting and API tools including Hugging Face Spaces and FastAPI. The transferable lesson is to extend a guided project independently and explain what each team member contributed.

Research Examples for STEM and Non-STEM Interests

Scientific research can begin with a narrow, testable question: comparing local water-quality measurements, analyzing public astronomy data, investigating patterns in environmental sensor readings, or evaluating model performance on an open dataset. Strong research involves a defined question, appropriate methods, honest reporting of negative or mixed results, and careful acknowledgment of mentors. Publishing is not required for a project to demonstrate learning. A paper submitted to a journal is not the same as an accepted or published paper.

Rohin portfolio example - Near-Earth asteroid research: Rohin is developing machine-learning approaches to estimate asteroid diameters and reduce uncertainty, with guidance from a Stanford graduate who works as a computer vision engineer. He has described comparisons involving Random Forest, XGBoost, and physics-based models, alongside a paper in progress. This is an example of mentored student-led research; the appropriate description depends on who formulated the question, wrote code, interpreted results, and authored the paper.

Humanities and social science students can conduct oral-history interviews with consent, study local history through archival records, analyze public transportation accessibility, or create a bilingual community resource. A student interested in psychology might review published research and develop an ethical survey only after following school and applicable research rules. Research quality depends on clear methods and appropriate safeguards, not on using advanced terminology.

Leadership, School Clubs, and Student-Led Initiatives

Leadership can be shown by planning events, onboarding new members, improving a club process, maintaining a shared resource, resolving scheduling problems, or supporting a team through a difficult project. Formal positions such as president, secretary, webmaster, or vice president can help describe responsibility, but the work accomplished matters more than the title. A student can also lead without an office by noticing an unmet need and organizing a response.

Rohin portfolio example - Gladiator Technology: Rohin serves as vice president of a school technology organization and reports more than 200 volunteer hours. He helps plan Media Center events, assists with schoolwide device distribution and collection, and supports students and staff. The broader lesson is that sustained operational responsibility and reliable service can be meaningful leadership, even when the work is not a competition or research award.

Rohin portfolio example - Future Business Leaders of America: Rohin helps maintain his chapter website with other webmasters, serves as an officer, and has competed in technology-related events. A student

with similar interests might maintain a robotics team website, publish a club newsletter, organize a showcase, or improve registration and communication workflows. Shared work should be described honestly rather than attributing the whole project to one person.

To start a new club or initiative, a student should first verify school policies, identify a genuine gap, talk with potential members and a faculty sponsor, set an achievable initial goal, and plan a small pilot. Examples include a coding circle for beginners, a peer writing workshop, an environmental data club, or a community repair event. Starting a club only for a leadership title, without actual participation or continuity, provides little meaningful benefit.

Examples Across Other Majors and Interests

Health and medicine: volunteer in permitted community health outreach, organize a first-aid awareness event with qualified adults, build accessible health-information materials using vetted sources, or shadow professionals where lawful and available. Avoid implying that a student is providing clinical advice or practicing medicine.

Business and economics: operate a small service business, track its finances, build an inventory tool for a school group, participate in entrepreneurship competitions, or study local consumer needs through interviews. A useful project explains the customer problem, ethical considerations, and what changed after feedback.

Engineering and robotics: join a robotics team, prototype an assistive device, improve a bicycle safety accessory, enter an engineering design challenge, or document a build-and-test cycle. Students should prioritize safe prototypes, teamwork, testing evidence, and lessons from failed designs.

Environmental science: coordinate a stream cleanup, perform a supervised biodiversity count, map local recycling access, analyze publicly available climate data, or help a community garden measure water use. Document actual outputs and avoid claiming environmental impact without measurements.

Writing, journalism, and history: edit a school publication, launch an interview podcast, research local historical landmarks, publish accessible explainers, or organize a student writing circle. Evidence may include published issues, interviews completed, editorial responsibilities, and reader feedback.

Arts, design, and media: create a portfolio, organize a student exhibition, compose music, produce a documentary, design accessible posters for nonprofits, or teach younger students creative skills. Growth can be shown through revisions, performances, commissions, and community collaboration.

Education and public service: tutor younger students, translate school resources where competent, design a study-skills workshop, organize a book drive, or mentor new club members. Track attendance, preparation, consistency, and student feedback without sharing minors private information.

Sports and athletics: compete, coach younger athletes under supervision, organize inclusive recreational activities, help manage team statistics, or lead safety and equipment initiatives. Athletic involvement can demonstrate discipline, teamwork, and commitment even without elite awards.

Competitions, Internships, and Accessible Alternatives

Competitions can provide deadlines, feedback, and a chance to collaborate. Examples include programming contests, science fairs, robotics competitions, business presentations, debate, writing competitions, and art showcases. Students should choose events that fit their interests and available preparation time. Participation and learning are valid outcomes; an award is not necessary to make the experience worthwhile.

Internships may be paid or unpaid and can involve research, software, design, communications, or operations. Students should verify supervision, age eligibility, safety, and compensation rules. A student without access to a formal internship can create a substantive independent project, contribute to a school or community organization, volunteer with an established nonprofit, or seek informational interviews. Access to internships varies greatly and should not be treated as a prerequisite for competitive college applications.

For example, a computer science student might contribute documentation to an open-source project, build a website for a community group with permission, or complete a self-directed software project with peer code review. A biology student might participate in a supervised citizen-science project; a prospective business major might analyze a school fundraiser and recommend improvements. These projects can demonstrate initiative when honestly documented.

Teaching, Mentorship, and Community Impact

Mentorship can connect academic strengths with community needs. A student might tutor math, help new students adjust to high school, teach beginner coding, mentor younger robotics participants, or run a workshop on finding extracurricular opportunities. Good teaching requires preparation, patience, and adapting explanations to learners rather than merely demonstrating personal expertise.

Rohin portfolio example - peer teaching: Rohin has taught beginner Python in a small camp, helped freshmen learn C in his computer science teachers classroom, and participated in SAT Math peer tutoring through Schoolhouse.world. These examples show several ways to share knowledge. Program eligibility or selection should be described separately from completed teaching sessions, and student counts should be reported only when confirmed.

To assess community impact, record concrete facts: how many sessions were held, how many people attended, whether participants returned, what materials were produced, and what feedback was received. Use privacy-preserving totals. Do not claim improved test scores, admissions outcomes, or community-wide benefits without evidence. A small project with sustained participation can be more meaningful than a large idea that never launches.

Choosing and Describing Activities Honestly

A student can select activities by asking: What problem or question interests me? Who could benefit? What resources do I actually have? Can I sustain this work without sacrificing sleep or core academics? What would meaningful progress look like in three months? Depth develops through repeated work, increasing skill, and reflection. Breadth can be useful for exploration, especially early in high school, but students need not pursue every available club.

When describing an activity, use a clear sequence: action taken, method or responsibility, outcome or evidence, and learning. Distinguish built from planned, submitted from published, selected from completed, and team contribution from independent work. A student can keep a simple activity log with dates, hours, role changes, artifacts, and confirmed results. There is no guaranteed extracurricular formula for selective college admission; authentic engagement and accurate reporting are the priority.